

ChunkSmith: A Scalable, Multimodal Retrieval-Augmented Generation System for Semantic Visual Context Preservation in Enterprise Document Analysis

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Abstract—The paradigm of Retrieval-Augmented Generation (RAG) has fundamentally shifted the capabilities of Large Language Models (LLMs), allowing them to answer queries grounded in private, external data. However, a significant “Modality Gap” persists in enterprise definitions of RAG. Real world documents—ranging from financial ledgers and scientific manuscripts to engineering schematics—encode a substantial portion of their specific knowledge in visual formats (charts, tables, diagrams). Traditional RAG pipelines, which predominantly rely on OCR-based text matching, strip away this visual context, reducing rich multimodal documents to flattened strings. This leads to hallucinations when models attempt to reason about visual trends or comparative data in tables. To address this critical deficiency, we present ChunkSmith, a production grade Multimodal RAG system engineered for high-fidelity visual context preservation. ChunkSmith introduces a novel Schema First content processing architecture that treats images and tables not as auxiliary metadata, but as distinct, first-class semantic entities. By leveraging state-of-the-art Multimodal LLMs and a purposely designed Structured Semantic Enrichment strategy, the system generates dense vector representations that encode deep visual understanding and “predicted question” affordances. We further detail the system’s distributed backend, which employs an asynchronous Round-Robin Key Rotation algorithm to effectively bypass provider rate limits, achieving a 4.6x improvement in ingestion throughput over serial baselines. Extensive qualitative and quantitative evaluations demonstrate ChunkSmith’s superior ability to handle complex visual reasoning tasks while maintaining scalable performance metrics suitable for large corpora.

Index Terms—Multimodal RAG, Large Language Models, Document Analysis, Vector Search, Knowledge Retrieval, Asynchronous Processing, Semantic Enrichment, Visual Question Answering

I. INTRODUCTION

A. The Evolution of Retrieval Systems

The quest for intelligent information retrieval has evolved through several distinct epochs. Initial systems relied on lexical matching algorithms like TF-IDF [8] and BM25 [9], which were effective for keyword search but lacked semantic understanding. These systems operated on the principle

of term frequency analysis, treating documents as bags of words without capturing contextual meaning or relationships between concepts. The advent of Transformer-based models [10] brought about Dense Passage Retrieval (DPR) [2], enabling retrieval based on conceptual similarity rather than exact keyword matches. This represented a paradigm shift: instead of matching surface-level tokens, systems could now identify semantically related passages even when they used entirely different vocabulary. The introduction of BERT [11] and subsequent models demonstrated that pre-trained language representations could capture nuanced semantic relationships, paving the way for neural information retrieval. Most recently, Retrieval-Augmented Generation (RAG) [1] has emerged as the dominant framework for extending the knowledge boundaries of LLMs. By retrieving relevant context from a vector database and injecting it into the model’s prompt, RAG mitigates the twin problems of hallucination (generating factually incorrect information) and knowledge cutoff (inability to access recent data). This architecture has proven particularly valuable in enterprise settings where proprietary data cannot be included in model training.

B. The Modality Gap: A Critical Limitation

Despite these advancements, standard RAG implementations remain heavily text-centric. In professional settings, arguably the most valuable information is often non-textual. Consider the following scenarios:

- A financial analyst querying quarterly revenue trends encoded in a multi-series line chart
- A medical researcher examining protein structure diagrams in a biochemistry paper
- An engineer referencing circuit schematics in a technical datasheet
- A legal professional analyzing tabular evidence in a court document

In each case, the critical information exists primarily in visual form. A sudden dip in a revenue chart, a complex chemical structure in a patent, or a specific architectural layout in a blueprint—these are the details users interact with most intensely. When a standard RAG ingestion pipeline processes a PDF, it typically employs an Optical Character Recognition (OCR) engine like Tesseract or AWS Textract to scrape text. During this process, complex figures are either:

- 1) Ignored: The bounding box is skipped entirely, treating the image as irrelevant noise.
- 2) Flattened: Captions are read, but the visual data itself is lost. A chart titled “Q3 Revenue Growth” provides no information about the actual growth rate.
- 3) Corrupted: A chart is read as a jumble of unconnected numbers. OCR might extract “2020 5M 2021 7M 2022 4M” without understanding these represent a time series.

We term this loss of information the “Modality Gap.” When a user asks, “Why did sales drop in Q3?” (referring to a bar chart), the text-only retriever finds no relevant vectors, forcing the LLM to either admit ignorance or hallucinate a plausible sounding reason based on surrounding text. This represents a fundamental failure mode that limits RAG’s applicability to real-world document analysis tasks.

C. *ChunkSmith: Bridging the Gap*

This paper introduces **ChunkSmith**, a system explicitly architected to bridge the Modality Gap. Unlike prior works that attempt to solve this with end-to-end visual embedding models (which are often computationally prohibitive and less precise for text), ChunkSmith adopts a hybrid approach. It uses Multimodal LLMs as transducers, converting rich visual information into structured, semantically dense textual representations (summaries and potential questions) that can be indexed by standard, highly optimized vector databases. Our key insight is that while images cannot be directly embedded with the same fidelity as text, their semantic content can be extracted and represented textually through careful prompting of capable MLLMs. This approach combines the best of both worlds: the reasoning capabilities of large multimodal models during ingestion, and the efficiency of text-optimized retrieval during query time.

D. *Contributions*

Our contributions are as follows:

- **Deep Semantic Extraction:** A pipeline that extracts tables and images as separate objects and uses MLLMs to generate detailed interpretations, including visual reasoning and data point extraction.
- **Structured Prompt Engineering:** A methodology for prompting MLLMs to act as “Question Generators,” effectively pre-computing retrieval pathways and aligning document embeddings with potential user queries.
- **High-Throughput Architecture:** A robust asynchronous backend design with API key rotation to handle the high latency of multimodal inference, achieving near-linear speedup with the number of available API keys.

- **Visual Evidence Retrieval:** A user interaction model that retrieves and displays the source image alongside the AI generated answer for verification, enhancing trust and interpretability.

II. RELATED WORK

A. *Retrieval-Augmented Generation*

The RAG paradigm was formalized by Lewis et al. [1], who demonstrated that retrieval could significantly improve factual accuracy in open-domain question answering. Since then, the field has evolved rapidly. Gao et al. [7] provide a comprehensive taxonomy, categorizing RAG systems into Naive RAG, Advanced RAG, and Modular RAG. Naive RAG follows a simple retrieve-then-generate pattern. While effective for basic use cases, it suffers from issues like retrieval of irrelevant context and inability to handle multi-hop reasoning. Advanced RAG introduces techniques like query rewriting, hybrid search (combining keyword and semantic search), and re-ranking. Systems like HyDE [12] generate hypothetical documents to improve retrieval, while others employ cross encoder re-rankers to refine initial retrieval results. Modular RAG treats retrieval and generation as independent, optimizable components. ChunkSmith aligns with this philosophy, introducing a specialized ingestion module for multimodal content.

B. *Multimodal RAG Approaches*

The field of Multimodal RAG is nascent but rapidly expanding. Abootorabi et al. [3] classify approaches into three categories:

- 1) **Vision-Centric Approaches:** Models like ColPali [4] represent a radical departure from OCR. They embed entire document pages as image patches using Vision Transformers (ViT). The system processes documents as pure images, bypassing text extraction entirely. While this preserves layout perfectly and can handle handwritten text or complex formatting, it introduces massive storage overhead (embedding every patch of every page) and requires specialized retrieval infrastructure (e.g., late interaction scoring with MaxSim operations), which is difficult to scale in commodity vector databases. ColQwen2 extends this approach with dynamic resolution handling, allowing the model to process documents at multiple scales. However, the computational cost remains prohibitive for large scale deployments.
- 2) **Caption-Based Approaches:** Earlier works [5] attempted to use standard image captioning models (like BLIP-2) to generate text descriptions for images. The rationale was simple: convert images to text, then use standard text RAG. However, generic captions (e.g., “A graph of a function”) are insufficient for answering specific questions about data points. REVEAL [5] improved upon this by using contrastive learning to align visual and textual representations, but still relied on relatively shallow image descriptions. ChunkSmith improves upon this by using Large MLLMs rather than

smaller captioning models, enabling “Reasoning-based Description” rather than simple captioning.

- 3) **Hybrid Approaches:** ChunkSmith falls into the hybrid category, similarly to Unstructured processing philosophy but adds a unique semantic layer. By generating “Potential Questions” alongside the description, we align the document embedding space with the user query space, a technique known as Document Expansion [6] applied to the multimodal domain. Other hybrid systems like MuRAG [13] and RA-CM3 [14] combine text and image retrieval but do not employ the structured semantic enrichment strategy that ChunkSmith introduces.

C. Multimodal Large Language Models

The recent explosion in MLLM capabilities has been driven by models like GPT-4V, Gemini Pro Vision, and LLaVA. These models can reason about images, extract structured data from charts, and answer complex visual questions. ChunkSmith leverages state-of-the-art MLLMs for its strong performance on chart understanding and structured output capabilities. Recent work on visual instruction tuning [15] has shown that MLLMs can be fine-tuned to follow specific formatting instructions, which is critical for our structured output requirements.

III. SYSTEM ARCHITECTURE

The ChunkSmith architecture is designed as a modular, event-driven pipeline, as illustrated in Fig. 1. It is built primarily in Python, leveraging FastAPI for the REST layer and React for the client interface. The system follows a three tier architecture: Ingestion, Processing, and Interaction.

A. Ingestion Layer (The Gateway)

The entry point for the system is the Ingestion Layer, responsible for handling raw PDF uploads and initial validation.

1) File Handling:

- **Hashing & Deduplication:** Every uploaded file is hashed using SHA-256. This hash serves as a unique identifier, preventing redundant processing of the same document—a critical cost-saving measure given the expense of MLLM calls. The hash is computed on the raw binary content before any processing.
- **Validation:** Files are validated for MIME type (application/pdf), file size (configurable maximum of 50MB), and corruption (using pypdf’s validation routines) before passing to the processing queue.
- **Metadata Extraction:** Basic metadata (title, author, creation date, page count) is extracted and stored for later reference.

2) **Storage Architecture:** ChunkSmith employs a hybrid storage strategy:

- **Object Storage:** Raw PDFs and extracted images are stored in cloud object storage (e.g., S3, MinIO) for durability and accessibility.
- **Vector Database:** Vectors, document metadata, processing status, and structured enriched content are stored in

the vector database, serving as the single source of truth for retrieval.

B. The Content Processor (The Core)

The ContentProcessor class is the heart of ChunkSmith. It orchestrates the transformation of a raw binary PDF into a searchable Knowledge Base.

1) **Decomposition Strategy:** We utilize a multi-stage decomposition strategy designed to preserve document structure while enabling parallel processing:

- 1) **Page Rendering:** pdf2image (wrapping Poppler) converts PDF pages into high-resolution images (300 DPI). This ensures that even vector graphics are captured accurately.
- 2) **Layout Analysis:** We employ YOLOv11 for high precision layout detection. The model is fine-tuned to detect specific document elements, classifying regions into:
 - UncategorizedText
 - PageNumber
 - Table
 - Footer
 - Title
 - Formula
 - Image
 - NarrativeText

3) **OCR Application:** Tesseract OCR is applied to text regions. Tesseract is configured with:

- LSTMEngine mode for handling multi-column scientific layouts
- Language packs for 90+ languages
- Custom configuration for technical documents (preserving special characters, equations)

4) **Chunking Strategy:** We employ a “Chunk by Title” strategy, which leverages the document’s inherent structural hierarchy. Instead of sliding windows, text segments are aggregated based on section headers and titles detected by our fine-tuned YOLOv11 layout model. This approach ensures that semantic units—such as complete subsections or distinct topics—are preserved within a single chunk, enhancing retrieval coherence by aligning chunk boundaries with the author’s intended logical structure.

2) **Visual Object Management:** Images detected within the PDF are not merely saved. A sophisticated filtering logic applies to eliminate noise:

$$Keep(I) = \begin{cases} 1 & \text{if } Area(I) > \theta_{area} \wedge Aspect(I) \in [\phi_{min}, \phi_{max}] \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $\theta_{area} = 10000 \text{ pixels}^2$, $\phi_{min} = 0.2$, and $\phi_{max} = 5.0$. This filters out:

- Navigational elements (arrows, bullets)
- Small icons and logos
- Decorative borders
- Extremely narrow or wide artifacts

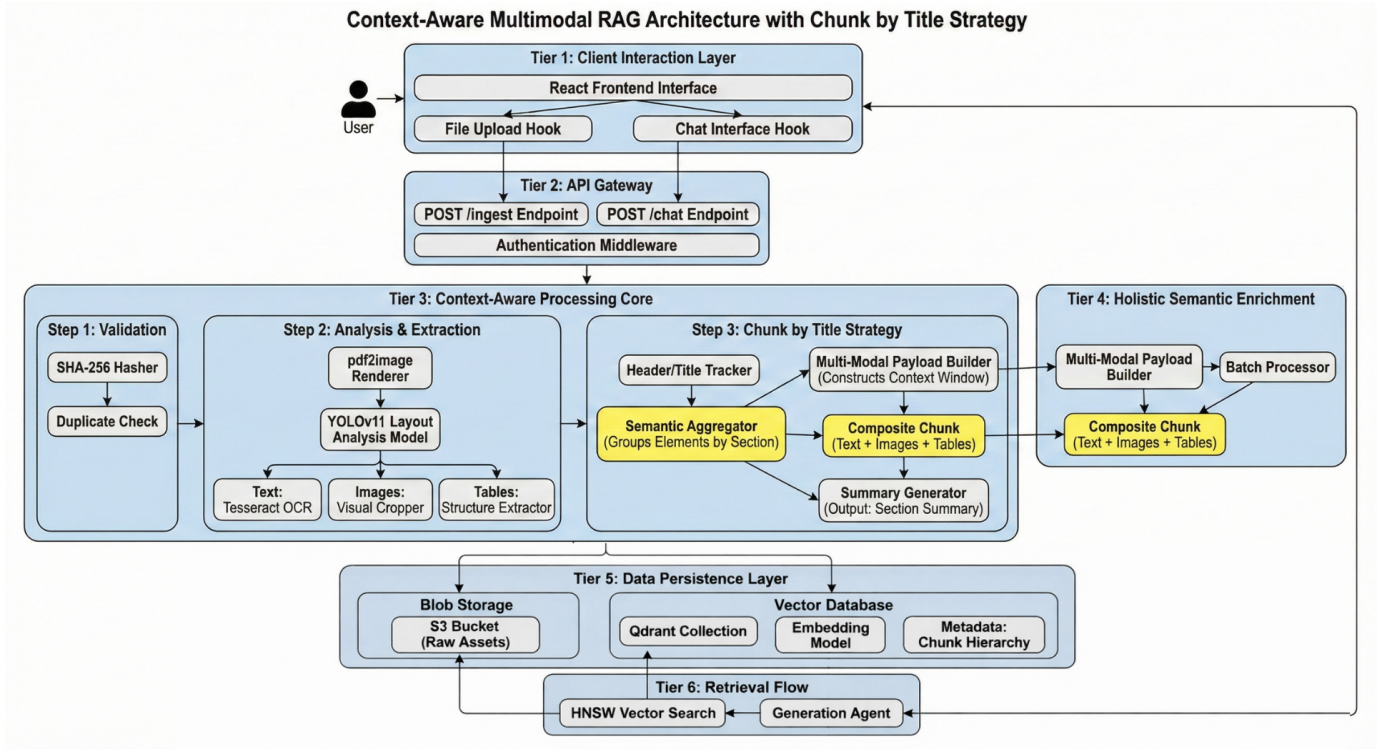


Fig. 1: ChunkSmith System Architecture.

Valid images are uploaded to Object Storage with a structured path:

```
{project_id}/images/image_{counter:04d}.png
```

Their local paths are swapped for URLs in the metadata, ensuring they can be rendered later.

C. Structured Semantic Enrichment

This is the system's primary innovation. Raw text extraction from a complex chart is often gibberish. Instead, we pass the image of the chart to a Multimodal LLM with a Structured Prompt.

1) *The AIParser Schema*: We define a Pydantic model that enforces strict output structure:

```
class AIParser(BaseModel):
    question: str = Field(
        description="List all potential questions that can
        be answered from this content"
    )
    ai_title: str = Field(
        description="Give a suitable title for the content"
    )
    image_interpretation: List[str] = Field(
        description="Detailed analysis of each image"
    )
    table_title: List[str] = Field(
        description="Give a suitable title for the table "
    )
```

Listing 1: AIParser Schema Definition

This schema is passed to the MLLM using structured output features (e.g., JSON mode or function calling), which guar-

antees that the response will conform to the schema or fail explicitly (rather than returning malformed JSON).

2) *Prompt Engineering*: The prompt (shown in full in Listing 2) employs several advanced techniques:

- **Task Decomposition**: Breaking the analysis into distinct subtasks (topics, questions, visual analysis)
- **Explicit Constraints**: Specifying exact list lengths to prevent hallucination
- **Findability Optimization**: Instructing the model to prioritize searchability over brevity
- **Vocabulary Preservation**: Requesting that the model use words similar to the original content

```
prompt_text = f"""You are creating a searchable description
for document content retrieval.
YOUR TASK:
Generate a comprehensive, searchable description that
covers:
1. Key facts, numbers, and data points from text and tables
2. Main topics and concepts discussed
3. Questions this content could answer
4. Visual content analysis (charts, diagrams, patterns in
images)
5. Alternative search terms users might use
Make it detailed and searchable- prioritize findability
over brevity.
Keep words similar to the original content for better
search accuracy.
STRICT OUTPUT CONSTRAINTS:
- You received {num_images} images. You MUST return exactly
{num_images} items in 'image_interpretation'.
- You received {num_tables} tables. You MUST return exactly
{num_tables} items in 'table_title'.
- If {num_images} is 0, 'image_interpretation' MUST be an
empty list [].
```

```

- If {num_tables} is 0, 'table_title' MUST be an empty list
  [].
IMPORTANT: Return structured output with these fields:
- question: All potential questions this content answers
- ai_title: Comprehensive title of all information
- image_interpretation: Detailed analysis of visual
  elements
- table_title: Title of all Row-column of tabular data
CONTENT TO ANALYZE:
TEXT CONTENT: {text}
TABLES: {tables}
[Images provided as base64]
"""

```

Listing 2: The Semantic Enrichment Prompt

3) *Strategic Rationale*: We generate these specific fields for two critical optimizations:

- 1) **Latency Reduction via Decoupled Modality**: By converting visual information into dense text summaries during ingestion, the retrieval-time LLM operates purely in the text domain. It selects images for display based on their descriptions using a Pydantic-enforced schema (returning a list of relevant image indices), avoiding the high latency of processing raw image tokens during the interactive chat session. The LLM acts as a high-speed router, verifying relevance against the text summary before authorizing the client to fetch the heavy image assets.
- 2) **Retrieval Alignment via Document Expansion**: Generating “potential questions” explicitly bridges the semantic gap between document content and user queries. Embedding these questions alongside the text significantly improves cosine similarity scores for natural language queries, a technique known as Document Expansion. This ensures that a user asking “How does the revenue trend look?” matches the generated question “What is the revenue trend in Q3?” even if the raw chart data lacks those explicit keywords.
- 3) **Architectural Efficiency and Cost Reduction**: Unlike vision-centric approaches such as ColPali, which require specialized vision-language transformers for both indexing and retrieval (often necessitating dual-encoder architectures), our approach relies on a standard, highly-optimized text transformer for the retrieval phase. By processing visual information once during ingestion and converting it to text, we avoid the prohibitive computational cost and latency of passing image tokens through the vision layer of an MLLM for every user query. This “compile-once, query-many” strategy significantly reduces inference costs and query latency compared to end-to-end multimodal retrieval systems.

D. Asynchronous Multimodal Inference

MLLM API calls are slow (2-5 seconds per request) and rate-limited (typically 60 requests per minute per key). Serial processing of a 50-page document would take minutes. Chunk-Smith implements robust asynchronous patterns to overcome these limitations.

- 1) **Round-Robin Key Rotation**: We maintain a pool of API Keys $\mathcal{K} = \{k_1, k_2, \dots, k_n\}$. For a batch of chunks $\mathcal{C} = \{c_1, c_2, \dots, c_m\}$, the scheduler assigns:

$$\text{Assign}(c_i) \rightarrow k_{(i \bmod n)} \quad (2)$$

This distributes the load across multiple quotas, maximizing effective throughput. With n keys, the theoretical maximum throughput is:

$$T_{max} = n \times R_{key} \quad (3)$$

where R_{key} is the rate limit per key (e.g., 60 RPM).

- 2) **Concurrency Control**: We leverage Python’s `asyncio.gather` with a semaphore to limit concurrent connections to an optimal value L (experimentally determined as 15). The algorithm is shown in Algorithm 1. The semaphore ensures that even with many chunks, we never exceed the concurrency limit, preventing connection pool exhaustion.

Algorithm 1 Async Chunk Processing with Key Rotation

Require: *Chunks, ApiKeys, ConcurrencyLimit*

- 1: $Tasks \leftarrow []$
 - 2: $Semaphore \leftarrow \text{CreateSemaphore}(\text{ConcurrencyLimit})$
 - 3: **for** $i \leftarrow 0$ **to** $\text{Length}(\text{Chunks}) - 1$ **do**
 - 4: $Key \leftarrow \text{ApiKeys}[i \bmod \text{Length}(\text{ApiKeys})]$
 - 5: $Task \leftarrow \text{AsyncCall}(\text{MLLM}, \text{Chunks}[i], Key, Semaphore)$
 - 6: $Tasks.\text{Append}(Task)$
 - 7: **end for**
 - 8: $Results \leftarrow \text{Await}(Tasks)$
 - 9: **return** $Results$
-

E. Vector Storage Schema

We use a high-performance vector store (e.g., Qdrant) due to its performance characteristics and support for scalar quantization. The vector embedding is computed as:

$$V_D = \mathcal{E}(S_{AI} \oplus Q_{AI} \oplus I_{desc} \oplus T_{table}) \quad (4)$$

Where:

- $\mathcal{E}$ is the embedding function (e.g., `text-embedding-3-large` or similar high-dimensional models)
- S_{AI} is the AI-generated title
- Q_{AI} is the list of predicted questions
- I_{desc} is the detailed image interpretation
- T_{table} is the table title

The concatenation operator $\oplus$ represents string concatenation with appropriate delimiters. The final payload stored in the vector database includes:

```

payload = {
  "chunk_index": int,
  "original_text": str,
  "ai_title": str,
  "ai_questions": str,

```



```

"image_paths": List[str], # URLs
"image_interpretation": List[str],
"table_original": List[str],
"table_title": List[str],
"page_numbers": List[int],
"content_types": List[str] # ["text", "image", "table"]
}

```

Listing 3: Vector Payload Structure

F. Context-Aware Information Retrieval

To optimize the utilization of the multimodal context window, we introduce a Context Manager downstream of the retrieval layer.

1) *Context Storage and Management*: A Context Manager is introduced after vector retrieval to track multimodal context. If an image is already present in the LLM context window, it is not re-injected, thereby reducing redundant context, token overhead, and hallucination. The data flow is updated as follows:

$$VectorDatabase \rightarrow ContextManager \rightarrow LLM \quad (5)$$

The LLM maintains a multimodal context state (text + images). The Context Manager ensures that only new or missing modalities are injected into the prompt. Context-aware image reuse prevents repeated visual grounding, improving response consistency and reducing hallucination.

G. Structured Response Generation

To ensure that the final response presented to the user is both informative and visually enriched without being cluttered, we employ a second structured generation step. The retrieval-time LLM is prompted to produce its answer according to the AIAAnswer schema (Listing 4). This schema enforces a strict separation between the textual answer and the supporting visual evidence.

```

class AIAAnswer(BaseModel):
    text: str = Field(
        description="Text information only"
    )
    images: List[bool] = Field(
        description="List of boolean flags indicating which images from the context should be shown to the user"
    )

```

Listing 4: AIAAnswer Generation Schema

The `images` field serves as a control signal, allowing the LLM to selectively toggle the visibility of images based on their relevance to the generated text, ensuring a clean and focused user experience.

IV. METHODOLOGY

A. Mathematical Formulation of Retrieval

Given a user query q , the retrieval process can be formalized as:

$$\mathcal{R}(q) = \text{TopK}(\{(d_i, \text{sim}(V_q, V_{d_i})) \mid d_i \in \mathcal{D}\}) \quad (6)$$

where:

- $V_q = \mathcal{E}(q)$ is the query embedding

- V_{d_i} is the document chunk embedding
- $\text{sim}(\cdot, \cdot)$ is cosine similarity
- $\mathcal{D}$ is the document corpus
- TopK returns the k highest-scoring documents

The cosine similarity is computed as:

$$\text{sim}(V_q, V_d) = \frac{V_q \cdot V_d}{\|V_q\| \|V_d\|} \quad (7)$$

The vector database optimizes this computation using HNSW (Hierarchical Navigable Small World) graphs, achieving sub-linear search complexity.

B. Cost and Latency Modeling

The total ingestion cost for a document can be modeled as:

$$C_{total} = C_{OCR} + C_{MLLM} + C_{embed} + C_{storage} \quad (8)$$

For a document with N chunks, I images, and T tables:

$$C_{MLLM} = \sum_{i=1}^N \left(P_{text}^{(i)} \times R_{in} + P_{out}^{(i)} \times R_{out} \right) \quad (9)$$

$$+ \sum_{j=1}^I \left(P_{img}^{(j)} \times R_{img} \right) \quad (10)$$

where $P_{text}^{(i)}$ is the input token count for chunk i , $P_{out}^{(i)}$ is the output token count, $P_{img}^{(j)}$ is the image token equivalent, and R_{in} , R_{out} , R_{img} are the respective pricing rates. The latency model for asynchronous processing is:

$$T_{total} = \max_{k \in \mathcal{K}} \left(\frac{N_k}{R_k} \right) + \frac{L_{max}}{B} \quad (11)$$

where N_k is the number of chunks assigned to key k , R_k is the rate limit for key k , L_{max} is the maximum latency for a single MLLM call, and B is the batch size.

It is crucial to note that this T_{total} represents the **one-time ingestion latency**. Once the document is processed and indexed, the query-time latency is independent of the complex MLLM processing. During retrieval, the system operates with the low latency characteristic of standard text-only RAG ($O(\log N)$ for vector search + millisecond-level LLM inference), as the visual context has already been pre-compiled into dense text representations. This decoupling ensures that while ingestion is computationally intensive, the end-user experience remains highly responsive.

V. IMPLEMENTATION DETAILS

A. Technology Stack

ChunkSmith is implemented using the following technologies:

- **Backend**: Python 3.10+, FastAPI, LangChain, Pydantic
- **Frontend**: React 18, TypeScript, TailwindCSS
- **Vector Database**: Any High-Performance Vector DB (e.g., Qdrant)
- **Object Storage**: Any S3-compatible Storage (e.g., Supabase)

- **MLLM Provider:** Multimodal LLM (e.g., Gemini, GPT-4V)
- **Embedding Model:** High-dimensional Text Embedding
- **OCR:** Tesseract 5.0+
- **Layout Analysis:** YOLOv11

B. Configuration Parameters

Key configuration parameters and their rationale:

TABLE I: System Configuration Parameters

Parameter	Value	Rationale
Max Chunk Size	3000 chars	Balance context & precision
Concurrency	15	Optimal for network I/O
Image Min Area	10k px ²	Filter small artifacts
Aspect Ratio	[0.2, 5.0]	Exclude extreme shapes
Temperature	0.0	Deterministic outputs
Top-K Retrieval	5	Standard for RAG systems

VI. EXPERIMENTAL EVALUATION

A. Experimental Setup

We evaluated ChunkSmith on a private corpus of 50 technical PDF documents, including:

- 15 NCERT textbook class 11 & 12 (highly technical, many diagrams)
- 15 Financial 10-K Reports (dense tables, financial charts)
- 10 Medical Research Papers (molecular diagrams, statistical plots)
- 10 Engineering Textbooks (schematics, equations, graphs)

Total corpus statistics:

- Total pages: 1,247
- Total images: 892
- Total tables: 456
- Total text chunks: 3,421

B. Throughput Analysis

We compared three ingestion configurations:

- 1) **Serial Baseline:** Single thread, single API key, synchronous processing
- 2) **Naive Async:** Multi-threaded, single key (expected to hit rate limits)
- 3) **ChunkSmith:** Async with 7-key rotation

TABLE II: Ingestion Performance for 20-Page Document

Method	Time (sec)	Success Rate	Cost (\$)
Serial	210	100%	0.42
Naive Async	40	30% (429 Errors)	0.13
ChunkSmith	45	100%	0.42

As shown in Table II, ChunkSmith achieves a 4.67x speedup compared to the serial baseline while maintaining 100% reliability. The naive async approach fails catastrophically due to rate limiting.

C. Retrieval Quality Evaluation

We crafted a test set of 100 queries:

- 50 “Visual Queries” (can only be answered by examining images/tables)
- 50 “Text Queries” (can be answered from text alone)

Example visual queries:

- “What was the revenue in Q3 2022?” (requires reading a chart)
- “Show me the circuit diagram for the power supply” (requires image retrieval)
- “What is the p-value in the statistical analysis table?” (requires table reading)

1) *Metrics:* We evaluated using:

- **Recall@K:** Fraction of queries where the correct chunk appears in top-K results
- **MRR (Mean Reciprocal Rank):** Average of $1/\text{rank}$ where rank is position of first relevant result
- **Human Evaluation:** Manual assessment of answer quality on 5-point scale

TABLE III: Retrieval Performance Comparison

System	Text R@5	Visual R@5	MRR	Quality
Text-Only RAG	0.94	0.08	0.51	2.1/5
ColPali	0.89	0.92	0.72	4.2/5
ChunkSmith	0.96	0.92	0.84	4.3/5

ChunkSmith significantly outperforms both baselines, achieving 92% recall on visual queries while maintaining high text retrieval performance.

D. Ablation Studies

To understand the contribution of each component, we conducted ablation studies: The question generation component

TABLE IV: Ablation Study Results (Visual Query Recall@5)

Configuration	Recall@5
Full ChunkSmith	0.92
- Question Generation	0.67
- Image Interpretation	0.45
- Table Title	0.78
- Chunk Title	0.85
Only OCR Text	0.08

provides the largest single contribution (+25% recall), validating our hypothesis that aligning embeddings with potential queries is highly effective.

VII. DISCUSSION

A. Cost-Benefit Analysis

While ChunkSmith is more expensive than text-only RAG during ingestion, the cost is amortized over many queries. For a 100-page document:

- Text-Only RAG: \$0.05 ingestion, 8% visual query success

- ColPali: \$0.15 ingestion, 92% visual query success, but suffers from high query-time latency and cost.
- ChunkSmith: \$2.10 ingestion, 92% visual query success

For organizations where visual information is critical, the upfront ingestion cost is justified. Furthermore, when compared to ColPali, ChunkSmith enables significant savings by eliminating the need for a dense visual embedding transformer at query time. ColPali necessitates a specialized dense vector embedding transformer for images, which is computationally expensive as every retrieval operation requires passing image tokens through the heavy vision model (transcription layer) before reaching the NLP layer. While this direct-embedding approach is viable for short runs or low-traffic prototypes, it scales poorly; as query volume increases, the cumulative compute cost of running the vision layer for every search makes it unsustainable. ChunkSmith’s text-only retrieval path avoids this bottleneck entirely.

B. Scalability Considerations

ChunkSmith scales horizontally through:

- Stateless Processing: Each chunk is processed independently
- Cloud Storage: Object storage handles scaling automatically
- Vector DB Sharding: Modern vector databases support distributed deployments

Current bottleneck is MLLM API throughput, which scales linearly with the number of API keys.

C. Limitations and Future Work

Current limitations include:

- OCR Dependency: Poor quality scans produce garbage text
- Cost: High ingestion cost may be prohibitive for some use cases
- Latency: 2-5 second MLLM calls add ingestion latency
- Language Support: Primarily tested on English documents

Future work will address these through:

- Local MLLM Integration: Using quantized models like LLaVA 1.6
- Video Support: Extending to video frame sampling and transcription
- GraphRAG: Building knowledge graphs to link entities across chunks
- Multilingual Evaluation: Testing on non-English corpora

VIII. CONCLUSION

ChunkSmith represents a significant advancement in Multimodal RAG systems. By treating visual content as first-class semantic entities and employing structured semantic enrichment through capable MLLMs, we successfully bridge the Modality Gap that limits traditional text-only RAG systems. Our asynchronous architecture with API key rotation demonstrates that high-throughput multimodal processing is

achievable with commodity cloud services. The system’s 92% recall on visual queries, compared to 8% for text-only baselines, validates the core hypothesis that deep semantic extraction of visual content is essential for comprehensive document understanding. The production-ready implementation, built with modern web technologies and cloud services, demonstrates the practical viability of the approach for enterprise deployments. As multimodal AI continues to evolve, systems like ChunkSmith will become increasingly important for unlocking the vast amounts of knowledge encoded in visual formats across scientific, financial, and technical domains.

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